# From Geometry to a Real-Time Digital Twin of Rolling-Disc Notched Tendon-Driven Continuum Robots

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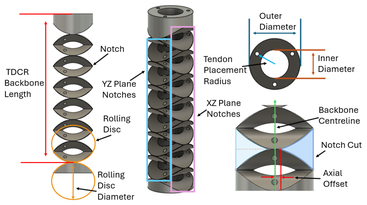
IITGN ROBOTICS

## PART A How does geometry govern compliance?

under review, ICRA 2027 — ICRA IEEE INTERNATIONAL CONFERENCE ON ROBOTICS AND AUTOMATION

### The robot, and two questions

Printed notched tubes are cheap continuum robots for confined spaces. Two things are missing: a rule for choosing the notch geometry, and a model that predicts the whole shape live, with no sensor on the robot.




**A. Design**
Which geometry gives a wanted bend, and what stroke and force does it cost?

**B. Model**
Can the whole shape be predicted live from servo commands alone, for any number of modules?

TPU tube, two orthogonal families of rolling-disc notches, tendons in the wall, stackable 160 mm modules.

**What is new**

- **A** SWEEP: geometry to FEM results with no manual step; 616 runs map how the notch geometry sets bend, stroke and force.
- **B** A twin built from the CAD's cross-sections, with pull-only tendons and gravity, solved exactly in 1 to 4 ms.
- **B** It learns the module's law, not the shape: fitted on one and two modules, it predicts four within 0.5 mm; a physics-informed network must be retrained.

### A1 SWEEP: geometry to simulation, unattended

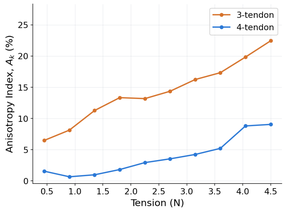
**Parameters** $D_{RD}$, $L$, $D_o$, $D_i$, $e$, $c$, $N_t$ → **CAD** CadQuery, one Boolean cut → **Mesh** CGAL tetrahedra → **FEM** SOFA, pull-only tendons → **Metrics** curvature, CC error, stroke




| | |
|---|---|
| designs | 14 |
| tendons × loads | 4 × 11 |
| FEM runs (done) | 616 (614) |
| $E$, TPU, measured | 56 MPa |
| gravity | on |
| claims valid for | $T \le 0.9$ N |

Same load, three disc diameters ($R_d = D_{RD}$): a larger disc gives more, shallower notches and a tighter bend.

### A2 Match the actuation to the structure

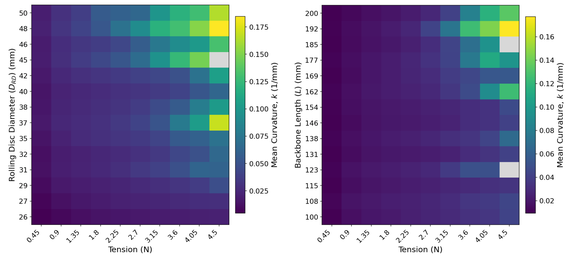



With three tendons the bend depends on its direction. The anisotropy index $A_\kappa = \sigma_\kappa / |\bar{\kappa}|$ falls from 6.2 % to 2.6 % with four tendons (13 of 14 designs).

Four tendons form two antagonistic pairs, so a section needs two motors.

Anisotropy index against tension for three and four tendons on one design.

### A3 What each parameter controls



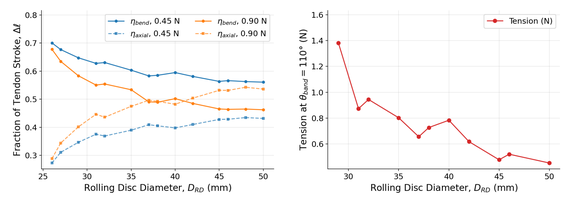

**$D_{RD}$ sets curvature** — turning 54° → 143° (26 to 50 mm, 0.9 N)

**$L$ sets reach** — tip travel 78 → 157 mm (100 to 200 mm)

Mean curvature (1/mm). More notches bend further, but deviate up to 5× more from constant curvature. The claims use $T \le 0.9$ N, the first two columns; grey cells have no settled run.

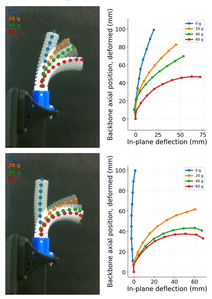
### A4 Stroke budget: what a bend costs the actuator

$$\underbrace{\Delta\ell}_{\text{tendon stroke}} = \underbrace{\Theta\, d}_{\text{bending}} + \underbrace{\Delta L}_{\text{axial compression}}$$




Left: the fractions of the stroke spent on bending and on compression. The budget closes to 0.99, and 42 to 47 % of the stroke shortens the body instead of bending it. Right: a larger disc lowers the tension for a 110° bend from 0.87 to 0.45 N.

### A5 Experiment: three printed prototypes




| $D_{RD}$ (mm) | NRMSE, % of length centreline | tip | axial |
|---|---|---|---|
| 30 | 18.5 | 20.9 | 43.1 |
| 67.89 | 10.1 | 23.5 | 11.9 |
| 86.84 | 16.3 | 64.7 | 23.9 |

**Shape agrees within 18.5 %**; the tip less well: friction and self-contact are not modelled. The prototypes span $D_{RD}$ = 30 to 87 mm, wider than the sweep.

## PART B A sensor-free, real-time digital twin

in preparation for IEEE RA-L — IEEE Robotics & Automation Society

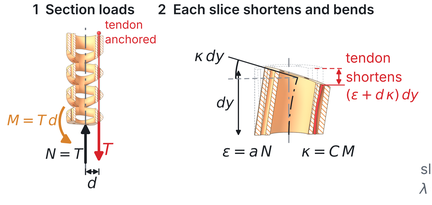
### B1 Servo commands in, whole shape out

**Servo commands** $\delta\ell$ → **Tendon map** $\delta\ell = G\lambda$ from the CAD → **Pull-only QP** $\lambda \ge 0$ exact slack → **Gravity** fixed point CAD weight → **Shape** every 5 mm in 1 to 4 ms

its weight moves the tendon paths

No sensor on the robot. The module is the SWEEP design with $D_{RD}$ = 30 mm, $L$ = 150 mm; one module model is composed by statics for any number of modules. 1 to 4 ms per pose on a laptop CPU (slowest 26 ms); an independent Cosserat rod takes about 360 ms.

### B2 The physics: a beam that follows the CAD

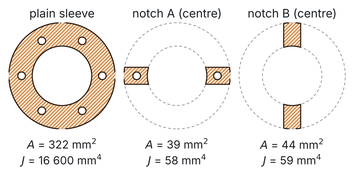



$$\delta\ell = G\lambda, \quad G_{ij} = \sum_c \tau_{ic}\tau_{jc}\left(a_c + \boldsymbol{d}_i^{\top} C_c\, \boldsymbol{d}_j\right) + c_t L_i\, \delta_{ij}$$

$$\lambda^{\star} = \arg\min_{\lambda \ge 0} \tfrac{1}{2}\lambda^{\top} G\lambda - \delta\ell^{\top}\lambda$$

**Tendon map:** exactly linear, any bend

**Pull-only:** convex, unique, exact



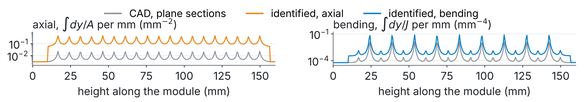

In a notch only two 4 mm bridges remain. The area falls eightfold and the bending stiffness about 280-fold, so the notches do the bending.

$\delta\ell$: servo displacements; $\lambda$: tendon tensions. $\tau_{ic} \in \{0, 1\}$: tendon $i$ passes slice $c$. $a_c = \int_c dy/A$, $C_c = \int_c dy/J$: slice compliances from the CAD. $c_t L_i$: tendon stretch.

### B3 Identify the module law once, then verify

| notch compliance | vs beam theory |
|---|---|
| axial | ×20.2 |
| bending, $x$ and $z$ | ×5.9, ×5.3 |
| torsion (twist test) | ×2.45 |

| solver check | difference |
|---|---|
| closed-form beam cases | < 1 µm |
| independent Cosserat rod | 0.004 mm |
| automated tests | 132 pass |




Slice compliance along one module: the CAD, and the law identified from 675 FEM poses of one and two modules (factors relative to this FEM).

### B4 Scientific ML: learn the law, not the solution

**Learned law** $\theta$ $C_\theta(y) = C_{CAD}(y)\, e^{\theta}$ 4 physical numbers → **Exact physics** pull-only QP + gravity → **Shape** → loss ← **Data** FEM now, mocap next

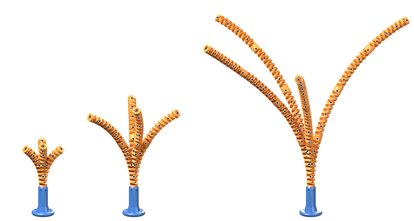
fit $\theta$ through the solver

**Next:** a convex notch energy $\psi_\theta(N, \boldsymbol{M})$, with $\theta$ a function of geometry (Direction, step 2)

| tip error vs FEM (mm) | 1 and 2 modules | 4 modules (unseen) |
|---|---|---|
| learn the **solution** (PINNs) | 3 to 6 | 54 to 73, retrained |
| **learn the law, in the solver** | **0.2 to 0.4** | **0.50, same $\theta$** |

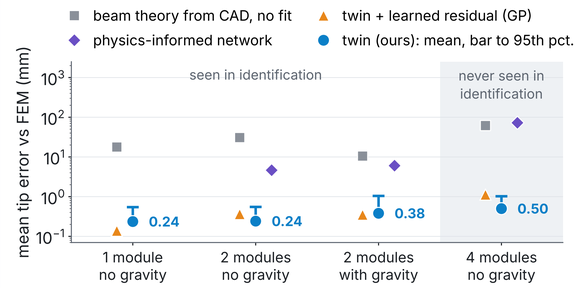
One $\theta$ for every number of modules. PINNs: forward and inverse, same data.

### B5 Calibrate small, deploy large (FEM ground truth)




The law is identified on one and two modules. On four, never used for it, the mean tip error is 0.50 mm (0.08 % of length), against 62 mm for beam theory from the CAD alone.

Renders: the twin on the CAD, at one scale. Dots: FEM centreline. Blue: mean tip error.




Mean and 95th percentile, 60 to 100 held-out poses per set, tip bends up to 75°. Coarse and fine FEM meshes differ by 0.6 mm at the tip (earlier robot): errors near 0.5 mm are at the FEM's resolution.

### B6 Towards hardware: what the data can identify

| problem (synthetic data) | remedy in the calibration |
|---|---|
| weight and stiffness confounded | weigh modules, fix the mass |
| equal pulls slacken distal tendons | pretension 1 : 1.75 : 2.25 : 2.5 |
| 0.4 mm moment-arm error | fit as 5th number, to 0.02 mm |



Right: the parts designed for the four-module robot.

## Conclusions

**A. Design.** The disc diameter sets curvature (54° → 143° at 0.9 N), the length sets reach, and 42 to 47 % of the stroke is compression.

**B. Model.** Identified on one and two modules, the twin predicts four to 0.50 mm (0.08 % of length) in 1 to 4 ms; a learned shape does not.

**Limits.** FEM only, at its resolution (about 0.6 mm); quasi-static; no friction or contact in the identification; bends up to 75°.

**Builds on:** Camarillo et al., T-RO 2008 • Rucker & Webster, T-RO 2011 • Faure et al., SOFA, 2012 • Coevoet et al., SoftRobots, 2017 • Kim et al., arXiv:2507.10121 • Danesh et al., arXiv:2608.26273.

## Direction A geometry-conditioned twin for inverse design

**1 Hardware validation** (now)
Motion capture, 4-module robot.
Calibrate **5 numbers on 15 poses** of 2 modules; test on 4 modules.
**Pass:** tip error ≤ 2 % of length (13 mm), fixed before any data.

→

**2 Law as a function of geometry**
SWEEP generates designs; the twin identifies each: $\theta = f(D_{RD}, D_i, L)$.
**Pass:** beats CAD-only theory on designs held out entirely.

→

**3 Inverse design**
Target curvature + stroke budget → geometry, in milliseconds; print and verify.
**If residuals demand:** convex notch energy $\psi_\theta$, friction, self-contact.

## Outputs and my role

**[1]** D. Yadav, **O. M. Pawar**, Krishnapriya R, S. Gandhi, M. Vadali, "How Geometry Governs Compliance? A Parametric Study of Rolling-Disc Notched Tendon-Driven Continuum Robot," **under review, IEEE ICRA 2027.**

**[2]** **O. M. Pawar** et al., sensor-free digital twin of a modular notched TDCR, in preparation for IEEE RA-L, with an installable open package.

**Role.** Part A: SOFA 25.06 port, parallel sweeps, mesh and settling checks. Part B: model, solver, identification, calibration, baselines and software.

**Thanks** to Divij Yadav, Krishnapriya R and Shaili Gandhi for SWEEP and the prototypes.

Report (PDF) Code (GitHub)